



Problem Statement

27× more users on one cell tower than another
under the exact same land-use label.

same place ≠ same people ≠ same network



JDSE 2026

11th Junior Conference on Data Sciences and Engineering

Beyond Static Cell Labels

Integrating Mobility and Performance Data for Fine-Grained Network Analysis

- Two real datasets · 1 week · Rio de Janeiro



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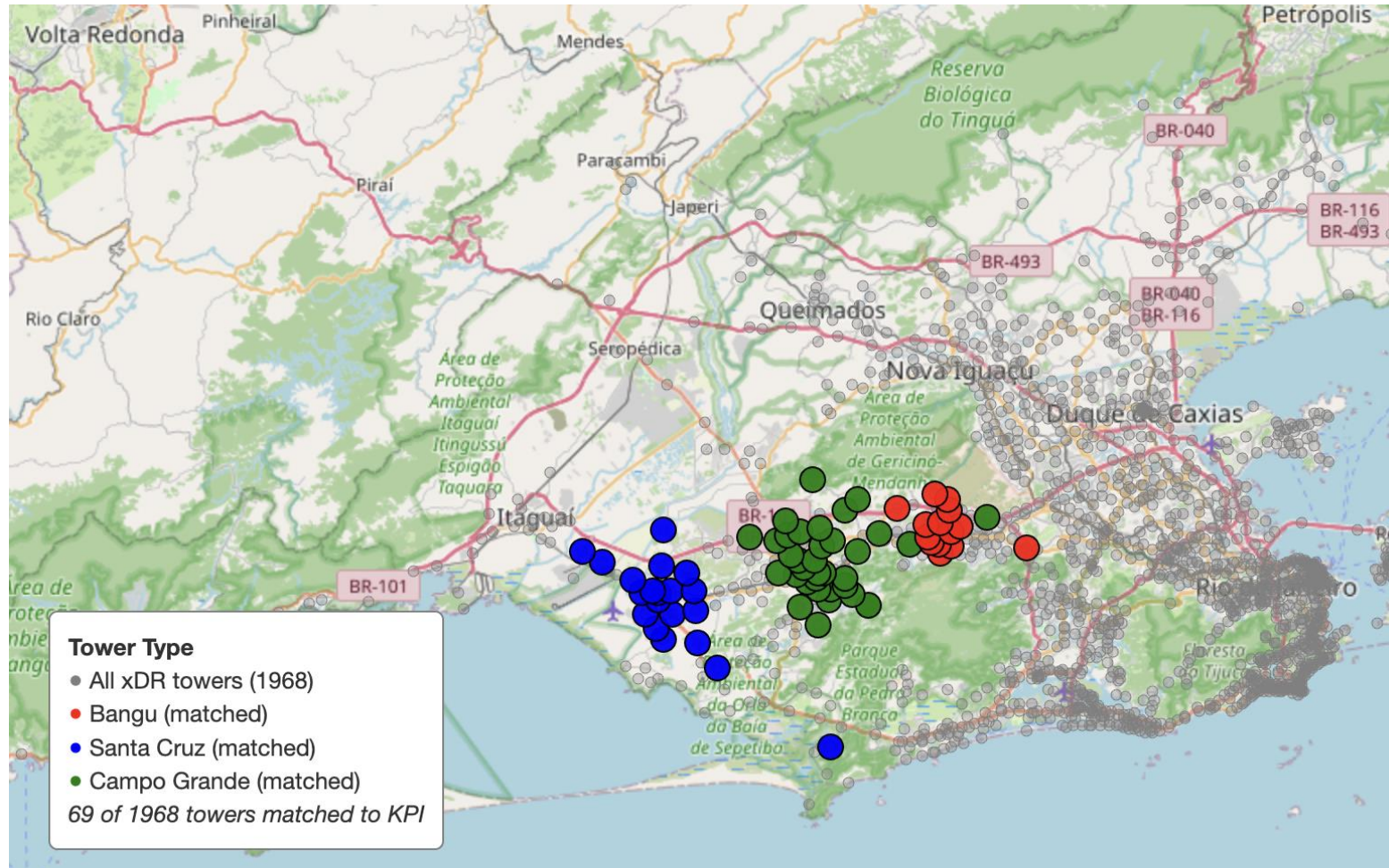
- Two real datasets · 1 week · Rio de Janeiro
- 5.3 billion mobility records matched with hourly network performance data



Beyond Static Cell Labels

Integrating Mobility and Performance Data for Fine-Grained Network Analysis

- Two real datasets · 1 week · Rio de Janeiro
- 5.3 billion mobility records matched with hourly network performance data
- Explored network load / visitation patterns



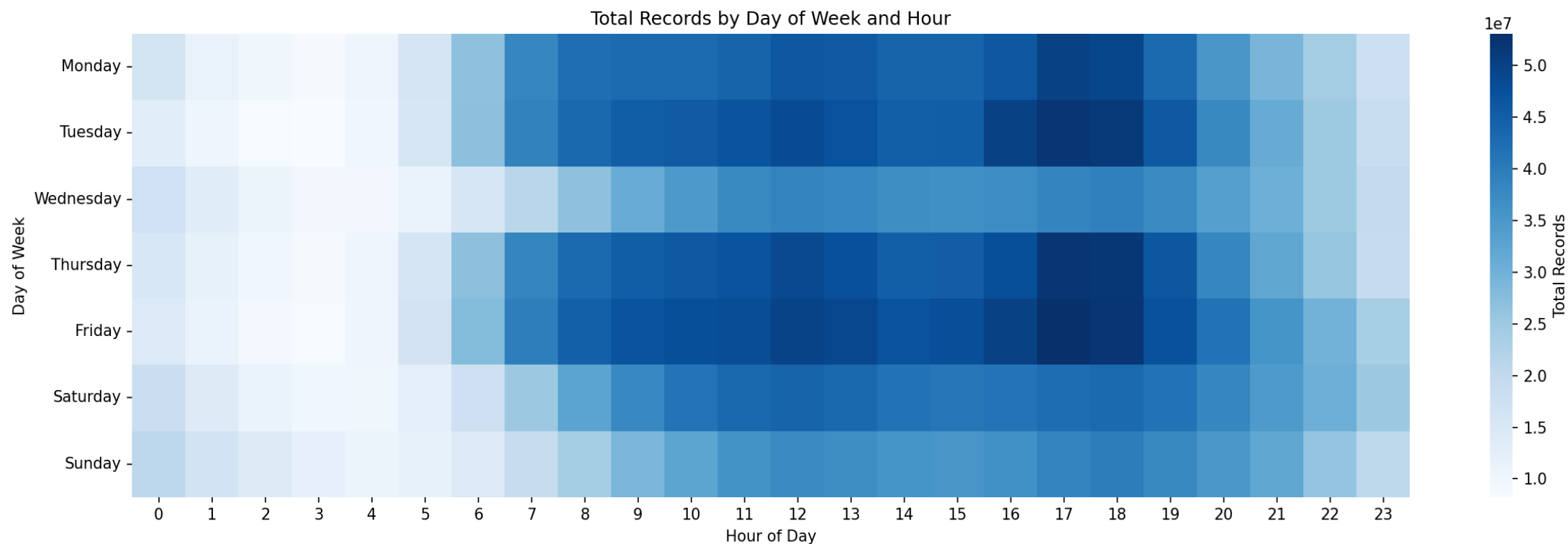
Spatial mapping between two datasets

For details, at **Poster [#27]** — come say hi.



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*Real weekly traffic pattern
from the study*

For details, at **Poster [#27]** — come say hi.

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